Answer key - Introduction to Differentiation

GCE Advanced Level - Mathématiques - Corrige

Detailed method for every exercise. Follow the substitutions line by line.
Corrige detaille de chaque exercice. Suis les substitutions ligne par ligne.

1. Differentiate f(x) = x^4.
Derive f(x) = x^4.

- 4x^3 *
- 4x^4
- x^3
- 3x^4

Answer: 4x^3
By the power rule, f'(x) = 4x^3.
Par la regle de puissance, f'(x) = 4x^3.

2. For f(x) = x^2, find the gradient at x = 3, that is the value of f'(3).
Pour f(x) = x^2, trouve la pente en x = 3, c est-a-dire la valeur de f'(3).

Answer: 6
f'(x) = 2x, so f'(3) = 6.
f'(x) = 2x, donc f'(3) = 6.

3. Differentiate f(x) = 3x^2 + 5x + 7.
Derive f(x) = 3x^2 + 5x + 7.

- 6x + 5 *
- 6x + 5 + 7
- 3x + 5
- 6x + 12

Answer: 6x + 5
Differentiate term by term: 3x^2 gives 6x, 5x gives 5, and the constant 7 gives 0. So f'(x) = 6x + 5.
On derive terme par terme : 3x^2 donne 6x, 5x donne 5, et la constante 7 donne 0. Donc f'(x) = 6x + 5.

4. Differentiate f(x) = x^6.
Derive f(x) = x^6.

- 6x^5 *
- 6x^6
- 5x^6
- x^5

Answer: 6x^5
By the power rule, f'(x) = 6x^5.
Par la regle de puissance, f'(x) = 6x^5.

5. For f(x) = x^3, find the gradient at x = 2, that is f'(2).
Pour f(x) = x^3, trouve la pente en x = 2, c est-a-dire f'(2).

Answer: 12
f'(x) = 3x^2, so f'(2) = 3 x 4 = 12.
f'(x) = 3x^2, donc f'(2) = 3 x 4 = 12.

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6. Differentiate f(x) = x^2 + 4x + 1.

Derive f(x) = x^2 + 4x + 1.

- 2x + 4 *
- 2x + 4 + 1
- 2x + 1
- x + 4

Answer: 2x + 4

Term by term: x^2 gives 2x, 4x gives 4, and 1 gives 0. So f'(x) = 2x + 4.

Terme par terme : x^2 donne 2x, 4x donne 4, et 1 donne 0. Donc f'(x) = 2x + 4.

7. In first principles, what does the chord gradient fraction approximate as h gets very small?

Dans les principes premiers, que devient la fraction de la pente de la corde lorsque h devient tres petit ?

- The gradient of the tangent, that is the derivative / La pente de la tangente, c est-a-dire la derivee *
- Zero for every function / Zero pour toutes les fonctions
- The value of f(x) itself / La valeur de f(x) elle-meme
- The y-intercept / L ordonnee a l origine

Answer: The gradient of the tangent, that is the derivative

As h tends to zero the chord becomes the tangent, so the fraction tends to the gradient of the curve at that point.

Quand h tend vers zero, la corde devient la tangente : la fraction tend vers la pente de la courbe en ce point.

8. What is the derivative of f(x) = 7 (a constant function)?

Quelle est la derivee de f(x) = 7 (fonction constante) ?

- 0 *
- 7
- 1
- 7x

Answer: 0

A constant function has a horizontal graph, so its gradient is zero everywhere. The derivative is 0.

Une fonction constante a une courbe horizontale : sa pente est nulle partout. La derivee vaut 0.

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